The rating held. The gap is economy entertainment.

A note for service leadership on Skytrax airline reviews. The figures are directional. Reviewers choose to write, and a year of ours in this file is about 50 reviews.

TLDR

Overall rating held. The gap to Singapore Airlines and Cathay Pacific lines up with economy in-flight entertainment, and that gap was already there in 2018-2019. The five steps below are the thought flow. Sections 3 to 7 document them. Section 8 is the recommendation in full, and section 9 is the SQL.

The five steps

Step 1 Overall rating held at 3.05 in 2023-2024 (105 reviews), from 3.04 in 2018-2019 (104 reviews). The move is 0.01, about 0.1 standard errors. Singapore is at 3.58 (224 reviews), a gap of 0.53, about 3.6 standard errors. Cathay is at 3.38 (110 reviews), a gap of 0.33, about 2.0 standard errors.

Step 2 Cabin mix does not carry that gap. Giving our cabin scores Singapore's mix of economy, business, and premium economy lifts the average from 3.05 to 3.09, a lift of 0.04. Economy is 76 of 105 recent reviews (72.4%) and rates 3.00, against Singapore economy 3.41 and Cathay economy 3.28.

Step 3 In-flight entertainment is the category below both benchmarks: 2.62 on 76 scores, against Singapore 3.93 and Cathay 3.94. Food is a smaller gap, and a smaller one versus Cathay than versus Singapore. Wifi is wide on 32 scores, 30.5% of reviews.

Step 4 The entertainment gap is in economy, and it is old. Economy entertainment is 2.64 on 55 scores, 1.28 below Singapore and 1.33 below Cathay, about 5.8 standard errors versus Singapore. In 2018-2019 the same cabin was 2.38 on 48 scores.

Step 5 The next cut does not move the number. Direct and connecting flights both sit near 2.6. Solo and couple leisure both sit near 2.6. No city pair and no aircraft model reaches 20 entertainment scores. The chain stops at the economy cabin.

Recommendation

Fund economy in-flight entertainment first. It is 2.64 on 55 scores in 2023-2024, 1.28 below Singapore and 1.33 below Cathay, and it was 2.38 in 2018-2019. Keep food on the list behind it. Count wifi, which covers 34.2% of economy reviews, before spending on it. Economy cabin staff is 3.64, ahead of Cathay at 3.21. Leave that category where it is.

Next step

Take the economy entertainment gap to onboard data before a purchase order. This extract leaves aircraft blank on 55 of 105 recent reviews, and no named airframe reaches 20 entertainment scores, so a fleet case needs a new extract. Read this file again when 2026 is a complete submitted year. The 2025 economy reading is 2.39 on 21 reviews, and it stays out of the gap until a second year confirms it.

1. How to read the rest of this note

1.1 The decision and the carriers

The reader is Vietnam Airlines. The decision is where the first service spend goes. Singapore Airlines and Cathay Pacific are the benchmark peers, shown one carrier at a time. Thai Airways is the regional peer, so a level rating has a neighbour. VietJet Air is the home-market carrier and stays out of the benchmark. A gap is a difference in level in the same window. A move is a difference for one carrier between windows.

1.2 The clock, the measures, and the floor

The clock is the submitted date, which is present on every review. The flown date is blank on about one in five of our reviews, so the windows use the date that is filled in. The baseline window is 2018-2019. The recent window is 2023-2024. The yearly table runs 2014-2025. 2026 is a partial year in this extract, dated 29 October 2026, and is off every table. 2020-2022 stay in the yearly table and stay out of the two-window comparison.

Overall rating is the mean of the category scores a reviewer actually filled in. It is already stored to two decimals on each review. A blank category is left out of that review's mean, so wifi, which is scored on a minority of reviews, pulls only the reviews that scored it. Recommend rate is the share of reviews that would recommend the airline. Figures below are shown to two decimals, and a gap is the difference of those displayed figures. A step needs at least 20 reviews in 2023-2024 in that cell. A thinner cell is reported and stays out of the chain. The language through the evidence is "this lines up with".

3. Overall rating held, and the benchmark gap is a level

3.1 The path, 2014-2025

Across the years in the table below, our overall rating sits under Singapore in every year and under Cathay in almost every year. VietJet Air is kept for the level comparison in 3.4, where the home-market gap can be read next to the benchmarks. 2021 is 4 of our reviews and 2022 is 15. Cathay in 2021 is 2 reviews at 1.00, and Thai in 2021 is 4 reviews at 1.73. Those thin years sit in the yearly table and stay outside both windows.

3.2 Our year, 2018-2025

Submitted year	Reviews	Overall rating	Recommend rate
2018	52	2.97	57.7%
2019	52	3.11	57.7%
2020	22	2.29	27.3%
2021	4	2.37	25.0%
2022	15	2.67	40.0%
2023	58	2.85	43.1%
2024	47	3.29	57.5%
2025	30	2.66	36.7%

2023 was 2.85 on 58 reviews. 2024 was 3.29 on 47. The recent-window figure of 3.05 averages a softer year and a firmer year. 2025 is 2.66 on 30 reviews. The cabin split for that year is the next table, and it stays out of the benchmark gap.

3.3 The 2025 watch, by cabin

Cabin, 2025	Reviews	Overall rating	Recommend rate	Floor
Economy Class	21	2.39	28.6%	in the watch
Business Class	5	3.82	80.0%	under 20, shown only here
Premium Economy	4	2.67	25.0%	under 20, shown only here

2025 overall is 2.66 on 30 reviews. Economy is the only cabin at the floor, at 2.39 on 21 reviews. Business and premium economy are under the floor.

3.4 The level in 2023-2024

Gap vs us is their rating minus ours. A positive number means they score higher.

Carrier	Role	2018-19 n	2018-19 rating	2023-24 n	2023-24 rating	Gap vs us	Recommend
Vietnam Airlines	Client	104	3.04	105	3.05	0.00	49.5%
Singapore	Benchmark	364	3.72	224	3.58	0.53	58.5%
Cathay	Benchmark	335	3.37	110	3.38	0.33	53.6%
Thai	Regional	171	3.38	112	2.99	-0.06	44.6%
VietJet	Home market	222	1.99	191	1.68	-1.37	13.1%

The gap to Singapore is 0.53 (about 3.6 standard errors). The gap to Cathay is 0.33 (about 2.0 standard errors). In the baseline, Singapore was 3.72 and Cathay was 3.37, against our 3.04. Singapore has come down from 3.72 to 3.58. The gap eased, and it is still half a point. Cathay held. Thai Airways moved from 3.38 to 2.99. We are level with Thai because their rating fell. VietJet Air is at 1.68 on 191 reviews. The home-market comparison is a different product, and it is a wide one in our favour. Recommend rate moved from 57.7% to 49.5%, about 1.2 standard errors. Singapore's recent rate is 58.5% and Cathay's is 53.6%. That move sits inside the noise of about 100 reviews a window, so the funding case stays with the level gap.

4. The gap is inside economy

4.1 Cabin mix in 2023-2024

Carrier	Reviews	Economy	Premium economy	Business	First
Vietnam Airlines	105	76 (72.4%)	7 (6.7%)	22 (20.9%)	0 (0%)
Singapore	224	124 (55.4%)	22 (9.8%)	75 (33.5%)	3 (1.3%)
Cathay	110	71 (64.5%)	10 (9.1%)	28 (25.4%)	1 (0.9%)
Thai	112	85 (75.9%)	2 (1.8%)	24 (21.4%)	1 (0.9%)

Economy is 72.4% of our recent reviews, 55.4% of Singapore's, and 64.5% of Cathay's. Singapore carries more business class (75 of 224). Applying Singapore's mix of economy, business, and premium economy to our own cabin scores moves our average from 3.05 to 3.09. The lift is 0.04. Singapore's first-class cell is 3 reviews and is left out of that reweight. Mix is a small piece of a half-point gap.

4.2 Rating by cabin, 2023-2024

Cabin	Vietnam Airlines	Singapore	Cathay	Gap vs Singapore	Gap vs Cathay
Economy	3.00 (n=76)	3.41 (n=124)	3.28 (n=71)	0.41	0.28
Business	3.43 (n=22)	3.77 (n=75)	3.61 (n=28)	0.34	0.18

Premium economy is 7 reviews at 2.41. It is under the floor, so it stays out of the chain. Business clears the floor on the rating (22 reviews at 3.43). The economy gap is the wider one, and it is three quarters of the file. The next question is which category inside that cabin.

4.3 Business recommend rate, kept out of the chain

Business recommend rate moved from 74.1% on 27 reviews in the baseline to 50.0% on 22 recently. The business overall rating held, 3.39 then and 3.43 now. Entertainment scores inside recent business are 15, under the floor. This cell is a separate watch. It does not identify the category the rest of the chain follows.

5. Entertainment is the category below both benchmarks

5.1 Seven category scores, all cabins, 2023-2024

Each figure is the mean among reviews that scored that category, with the number of scores in parentheses. The rank that matters is the gap to the nearer benchmark, so a category leads only when both peers sit above us.

Category	Ours (scores)	Singapore	Cathay	Vs Singapore	Vs Cathay
Seat comfort	3.22 (101)	3.62 (213)	3.39 (109)	-0.40	-0.17
Cabin staff	3.72 (100)	4.07 (212)	3.49 (109)	-0.35	0.23
Food and beverages	3.04 (90)	3.58 (206)	3.23 (104)	-0.54	-0.19
In-flight entertainment	2.62 (76)	3.93 (192)	3.94 (96)	-1.31	-1.32
Ground service	2.98 (104)	3.40 (218)	3.56 (110)	-0.42	-0.58
Wifi and connectivity	2.22 (32)	3.68 (127)	3.32 (50)	-1.46	-1.10
Value for money	3.18 (105)	3.33 (224)	3.11 (110)	-0.15	0.07

5.2 What the rank says

In-flight entertainment is 2.62 on 76 scores, against Singapore 3.93 and Cathay 3.94. Both gaps are above a point. Wifi is 2.22 on 32 scores, against 3.68 and 3.32. The wifi gap is wide, and the score exists on 30.5% of our recent reviews. Entertainment is scored on 72.4%.

Food is 3.04 on 90 scores. The gap is 0.54 versus Singapore and 0.19 versus Cathay. Ground service is 2.98 on 104 scores, 0.42 below Singapore and 0.58 below Cathay. Inside economy that Singapore ground gap closes: economy ground is 3.03 on 75 scores, against Singapore 3.11 and Cathay 3.42.

Cabin staff is 3.72 on 100 scores. That is 0.23 ahead of Cathay and 0.35 behind Singapore. Value for money is 3.18, level with Singapore at 3.33 and Cathay at 3.11. On the trip as a paid fare, we sit with Cathay.

6. The entertainment gap is in economy, and it is old

6.1 Economy entertainment in both windows

Window	Vietnam Airlines	Singapore	Cathay
2018-2019	2.38 (48)	3.94 (222)	3.66 (186)
2023-2024	2.64 (55)	3.92 (105)	3.97 (61)

Scores, not reviews. The count in parentheses is the number of economy reviews that scored entertainment.

Recent economy entertainment is 2.64 on 55 scores, 72.4% of economy reviews. Singapore economy is 3.92 on 105 scores. Cathay economy is 3.97 on 61 scores. The gap versus Singapore is 1.28, about 5.8 standard errors.

In 2018-2019 the same cabin was 2.38 on 48 scores, against Singapore 3.94 and Cathay 3.66. The Singapore gap moved from 1.56 to 1.28. The Cathay gap moved from 1.28 to 1.33. Our economy score rose. The level versus both benchmarks is still about a point and a quarter. This is a standing gap, and the recent window did not open it.

6.2 The same cut for food, so the comparison is explicit

Economy food is 2.98 on 63 scores, against Singapore 3.46 (111 scores) and Cathay 3.24 (66 scores). The gaps are 0.48 and 0.26. Food is a real Singapore gap and a smaller Cathay gap. It is about a third of the entertainment gap versus the nearer peer. Economy cabin staff is 3.64 on 72 scores, ahead of Cathay at 3.21 and 0.29 behind Singapore at 3.93.

6.3 Business entertainment stays under the floor

Business entertainment is 2.73 on 15 scores. Singapore business is 3.98 on 63 scores. Cathay business is 4.00 on 25 scores. The point estimate sits in the same place as economy. The cell has 15 scores, so it does not enter the chain and it does not support a separate business-cabin case from this file.

7. The next cut does not move the number

A cell enters this table's chain column when it has at least 20 entertainment scores in 2023-2024. The question is whether entertainment then leaves 2.6.

Cut, 2023-2024	Reviews	Overall rating	Entertainment	Scores	Chain
Direct	77	3.15	2.62	53	yes, same level
Layover	28	2.78	2.61	23	yes, same level
Solo Leisure	41	3.24	2.62	29	yes, same level
Couple Leisure	31	3.23	2.67	24	yes, same level
Family Leisure	22	2.71	2.47	15	under 20 scores
Business	11	2.52	2.75	8	under 20 scores
Verified	87	3.14	2.73	59	yes, same level
Not verified	18	2.61	2.24	17	under 20 scores

7.1 Connection and traveller type

Direct flights score entertainment at 2.62 on 53 scores. Flights with a layover score it at 2.61 on 23 scores. Solo leisure is 2.62 on 29 scores. Couple leisure is 2.67 on 24 scores. The level does not gather in connections, and it does not gather in one of the two leisure cells large enough to read.

Family leisure is the weak cell on overall rating: 2.71 on 22 reviews, against Singapore 3.38 (49) and Cathay 3.14 (24). Entertainment scores in that cell are 15, under the floor. Solo leisure, the larger cell, is already at 3.24 overall against Singapore 3.84 and Cathay 3.64, and its entertainment score is 2.62. Family is a later look. It is not the step that explains entertainment.

7.2 Verified reviews

Verified reviews are 87 of 105. Entertainment on them is 2.73 on 59 scores. Not-verified reviews are 18, under the floor. The gap is present in the verified majority.

7.3 Route and aircraft

The busiest city pair has 6 reviews (Sydney (SYD) to Paris (CDG)). The next pairs are Hanoi (unknown) to Ho Chi Minh City (unknown) (5); Bangkok (BKK) to Ho Chi Minh City (unknown) (3); Hanoi (unknown) to Hue (unknown) (3); Hue (unknown) to Hanoi (unknown) (2). None of them reach 20 reviews. The origin airport code is blank on 57 of 105 reviews, and the destination code is blank on 63. A route case is not available in this extract.

Aircraft is blank on 55 of 105 reviews. unknown: 55 reviews, entertainment 2.43 on 37 scores. Airbus A321: 22 reviews, entertainment 2.31 on 13 scores. Airbus A350: 14 reviews, entertainment 3.07 on 14 scores. Boeing 787: 7 reviews, entertainment 3.17 on 6 scores. Boeing 787-9: 3 reviews, entertainment 3.67 on 3 scores. Airbus A320: 2 reviews, entertainment 1.00 on 1 scores. Boeing 787-10: 1 reviews, entertainment 4.00 on 1 scores. Airbus A321neo: 1 reviews, entertainment 1.00 on 1 scores. Every named airframe is under 20 entertainment scores. The point estimates on the A350 and the A321 differ, and this file cannot carry a fleet case. The chain stops at the economy cabin.

8. What to fund

Put the next service spend on economy in-flight entertainment. It is 2.64 on 55 scores in 2023-2024, 1.28 below Singapore and 1.33 below Cathay, and it was 2.38 in 2018-2019.

8.1 Economy entertainment

The cell is large enough to act on (55 scores, 72.4% of economy reviews). It lines up with the overall gap after mix, cabin, and category are peeled off. It shows up on direct and connecting flights, and on solo and couple leisure. The standard error of the economy gap versus Singapore is about 0.22 points.

8.2 Food stays behind it

Economy food is 2.98 on 63 scores. That is 0.48 below Singapore and 0.26 below Cathay. Keep it on the list, behind entertainment. The nearer benchmark is a quarter of a point away. Entertainment is more than a point away.

8.3 Count wifi before spending on it

Economy wifi is 2.23 on 26 scores, against Singapore 3.55 and Cathay 3.28. The shape matches entertainment. The score covers 34.2% of economy reviews. Raise completion of that score, in the survey or in the next extract, before a wifi business case. Thirty-two scores across all cabins is enough to see the shape and too few to price a fix.

8.4 Keep cabin staff where it is

Economy cabin staff is 3.64 on 72 scores. Cathay economy is 3.21. Singapore economy is 3.93. This is the category that already beats one benchmark peer. A spend plan that trades crew attention for a screen would spend the category we can point to.

8.5 What would reopen the decision

Reopen this file when 2026 is a complete submitted year. The 2025 economy reading is 2.39 on 21 reviews. A second year at that level, with at least 20 economy reviews, would put the overall rating back on the table beside entertainment. Reopen the fleet question with an extract that fills in aircraft: this one leaves it blank on 55 of 105 recent reviews, and no named airframe reaches 20 entertainment scores. Reopen wifi if a later window scores it on a majority of economy reviews and the gap holds.

The source is self-selected Skytrax reviews. A fix that moves the passengers who write reviews can move these figures. A fix that moves everyone else can be invisible here. Use the gap as the place to look, and check it against onboard data before the purchase order.

9. SQL

DuckDB, read-only, against the local mart. Each query is one of the statements this briefing runs. Schemas: airlines_reviews.fct_airline_review, conformed.dim_airline, conformed.dim_aircraft, airlines_reviews.dim_location. Overall rating is the review-level average_rating. Category means ignore blank scores because avg does.

9.1 Yearly overall rating and recommend rate, sections 3.1 and 3.2

```
SELECT
    a.airline_name_cleaned AS airline_key,
    year(f.date_submitted_id) AS y,
    count(*) AS n,
    avg(f.average_rating) AS rating,
    stddev_samp(f.average_rating) AS rating_sd,
    avg(CASE WHEN f.recommended THEN 1.0 ELSE 0.0 END) AS rec_rate
FROM airlines_reviews.fct_airline_review AS f
JOIN conformed.dim_airline AS a USING (airline_id)
WHERE a.airline_name_cleaned IN (
    'vietnam airlines', 'singapore airlines', 'cathay pacific airways',
    'thai airways', 'vietjet air'
)
  AND year(f.date_submitted_id) BETWEEN 2014 AND 2025
  AND f.average_rating IS NOT NULL
GROUP BY 1, 2
ORDER BY 1, 2
```

9.2 Two windows, overall rating and the seven categories, sections 3.4 and 5

```
SELECT
    a.airline_name_cleaned AS airline_key,
    CASE
        WHEN year(f.date_submitted_id) BETWEEN 2018 AND 2019 THEN 'baseline'
        WHEN year(f.date_submitted_id) BETWEEN 2023 AND 2024 THEN 'recent'
    END AS period,
    count(*) AS n,
    avg(f.average_rating) AS rating,
    stddev_samp(f.average_rating) AS rating_sd,
    avg(CASE WHEN f.recommended THEN 1.0 ELSE 0.0 END) AS rec_rate,
    avg(f.seat_comfort) AS seat,
    count(f.seat_comfort) AS n_seat,
    avg(f.cabin_staff_service) AS cabin,
    count(f.cabin_staff_service) AS n_cabin,
    avg(f.food_and_beverages) AS food,
    count(f.food_and_beverages) AS n_food,
    avg(f.inflight_entertainment) AS ife,
    count(f.inflight_entertainment) AS n_ife,
    avg(f.ground_service) AS ground,
    count(f.ground_service) AS n_ground,
    avg(f.wifi_and_connectivity) AS wifi,
    count(f.wifi_and_connectivity) AS n_wifi,
    avg(f.value_for_money) AS vfm,
    count(f.value_for_money) AS n_vfm
FROM airlines_reviews.fct_airline_review AS f
JOIN conformed.dim_airline AS a USING (airline_id)
WHERE a.airline_name_cleaned IN (
    'vietnam airlines', 'singapore airlines', 'cathay pacific airways',
    'thai airways', 'vietjet air'
)
  AND year(f.date_submitted_id) IN (2018, 2019, 2023, 2024)
  AND f.average_rating IS NOT NULL
GROUP BY 1, 2
ORDER BY 1, 2
```

9.3 Cabin, including 2025, sections 4 and 6

```
SELECT
    a.airline_name_cleaned AS airline_key,
    CASE
        WHEN year(f.date_submitted_id) BETWEEN 2018 AND 2019 THEN 'baseline'
        WHEN year(f.date_submitted_id) BETWEEN 2023 AND 2024 THEN 'recent'
        WHEN year(f.date_submitted_id) = 2025 THEN 'y2025'
    END AS period,
    f.seat_type,
    count(*) AS n,
    avg(f.average_rating) AS rating,
    stddev_samp(f.average_rating) AS rating_sd,
    avg(CASE WHEN f.recommended THEN 1.0 ELSE 0.0 END) AS rec_rate,
    avg(f.food_and_beverages) AS food,
    count(f.food_and_beverages) AS n_food,
    avg(f.inflight_entertainment) AS ife,
    count(f.inflight_entertainment) AS n_ife,
    stddev_samp(f.inflight_entertainment) AS ife_sd,
    avg(f.seat_comfort) AS seat,
    count(f.seat_comfort) AS n_seat,
    avg(f.cabin_staff_service) AS cabin,
    count(f.cabin_staff_service) AS n_cabin,
    avg(f.ground_service) AS ground,
    count(f.ground_service) AS n_ground,
    avg(f.wifi_and_connectivity) AS wifi,
    count(f.wifi_and_connectivity) AS n_wifi,
    avg(f.value_for_money) AS vfm,
    count(f.value_for_money) AS n_vfm
FROM airlines_reviews.fct_airline_review AS f
JOIN conformed.dim_airline AS a USING (airline_id)
WHERE a.airline_name_cleaned IN (
    'vietnam airlines', 'singapore airlines', 'cathay pacific airways',
    'thai airways'
)
  AND year(f.date_submitted_id) IN (2018, 2019, 2023, 2024, 2025)
  AND f.average_rating IS NOT NULL
GROUP BY 1, 2, 3
ORDER BY 1, 2, 3
```

9.4 Connection, traveller, verified trip, and aircraft, section 7

```
SELECT
    CASE
        WHEN f.has_layover THEN 'Layover'
        ELSE 'Direct'
    END AS cut,
    'connection' AS family,
    count(*) AS n,
    avg(f.average_rating) AS rating,
    avg(f.inflight_entertainment) AS ife,
    count(f.inflight_entertainment) AS n_ife
FROM airlines_reviews.fct_airline_review AS f
JOIN conformed.dim_airline AS a USING (airline_id)
WHERE a.airline_name_cleaned = 'vietnam airlines'
  AND year(f.date_submitted_id) BETWEEN 2023 AND 2024
  AND f.average_rating IS NOT NULL
GROUP BY 1
UNION ALL
SELECT
    f.type_of_traveller,
    'traveller',
    count(*),
    avg(f.average_rating),
    avg(f.inflight_entertainment),
    count(f.inflight_entertainment)
FROM airlines_reviews.fct_airline_review AS f
JOIN conformed.dim_airline AS a USING (airline_id)
WHERE a.airline_name_cleaned = 'vietnam airlines'
  AND year(f.date_submitted_id) BETWEEN 2023 AND 2024
  AND f.average_rating IS NOT NULL
GROUP BY 1
UNION ALL
SELECT
    CASE WHEN f.is_verified THEN 'Verified' ELSE 'Not verified' END,
    'verified',
    count(*),
    avg(f.average_rating),
    avg(f.inflight_entertainment),
    count(f.inflight_entertainment)
FROM airlines_reviews.fct_airline_review AS f
JOIN conformed.dim_airline AS a USING (airline_id)
WHERE a.airline_name_cleaned = 'vietnam airlines'
  AND year(f.date_submitted_id) BETWEEN 2023 AND 2024
  AND f.average_rating IS NOT NULL
GROUP BY 1
UNION ALL
SELECT
    ac.aircraft_model,
    'aircraft',
    count(*),
    avg(f.average_rating),
    avg(f.inflight_entertainment),
    count(f.inflight_entertainment)
FROM airlines_reviews.fct_airline_review AS f
JOIN conformed.dim_airline AS a USING (airline_id)
JOIN conformed.dim_aircraft AS ac USING (aircraft_id)
WHERE a.airline_name_cleaned = 'vietnam airlines'
  AND year(f.date_submitted_id) BETWEEN 2023 AND 2024
  AND f.average_rating IS NOT NULL
GROUP BY 1
ORDER BY 2, 3 DESC
```

9.5 Busiest city pairs, section 7.3

```
SELECT
    o.city || ' (' || o.airport || ') to '
        || d.city || ' (' || d.airport || ')' AS route,
    count(*) AS n,
    avg(f.average_rating) AS rating,
    avg(f.inflight_entertainment) AS ife,
    count(f.inflight_entertainment) AS n_ife
FROM airlines_reviews.fct_airline_review AS f
JOIN conformed.dim_airline AS a USING (airline_id)
JOIN airlines_reviews.dim_location AS o
    ON f.origin_location_id = o.location_id
JOIN airlines_reviews.dim_location AS d
    ON f.destination_location_id = d.location_id
WHERE a.airline_name_cleaned = 'vietnam airlines'
  AND year(f.date_submitted_id) BETWEEN 2023 AND 2024
  AND f.average_rating IS NOT NULL
GROUP BY 1
ORDER BY n DESC
LIMIT 5
```

9.6 Blank airport codes and blank aircraft, section 7.3

```
SELECT
    count(*) AS n,
    count(*) FILTER (WHERE o.airport = 'unknown') AS origin_unknown,
    count(*) FILTER (WHERE d.airport = 'unknown') AS dest_unknown,
    count(*) FILTER (WHERE ac.aircraft_model = 'unknown') AS aircraft_unknown
FROM airlines_reviews.fct_airline_review AS f
JOIN conformed.dim_airline AS a USING (airline_id)
JOIN airlines_reviews.dim_location AS o
    ON f.origin_location_id = o.location_id
JOIN airlines_reviews.dim_location AS d
    ON f.destination_location_id = d.location_id
JOIN conformed.dim_aircraft AS ac USING (aircraft_id)
WHERE a.airline_name_cleaned = 'vietnam airlines'
  AND year(f.date_submitted_id) BETWEEN 2023 AND 2024
  AND f.average_rating IS NOT NULL
```

9.7 Leisure cells for the benchmark peers, section 7.1

```
SELECT
    a.airline_name_cleaned AS airline_key,
    f.type_of_traveller,
    count(*) AS n,
    avg(f.average_rating) AS rating,
    avg(f.inflight_entertainment) AS ife,
    count(f.inflight_entertainment) AS n_ife
FROM airlines_reviews.fct_airline_review AS f
JOIN conformed.dim_airline AS a USING (airline_id)
WHERE a.airline_name_cleaned IN (
    'vietnam airlines', 'singapore airlines', 'cathay pacific airways'
)
  AND year(f.date_submitted_id) BETWEEN 2023 AND 2024
  AND f.average_rating IS NOT NULL
  AND f.type_of_traveller IN ('Solo Leisure', 'Couple Leisure', 'Family Leisure')
GROUP BY 1, 2
ORDER BY 2, 1
```